

10. Define Supervised Learning

Definition:

Supervised learning is a type of machine learning where the model is trained using labeled data.

Explanation:

The input data already has correct answers (output), and the model learns to map input → output.

Example:

Email spam detection (spam / not spam)

11. Define Unsupervised Learning

Definition:

Unsupervised learning is a type of learning where the model is trained using unlabeled data.

Explanation:

The system tries to find patterns or groups in data on its own.

Example:

Customer segmentation (grouping customers based on behavior)

12. What is Reinforcement Learning?

Definition:

Reinforcement learning is a learning method where an agent learns by interacting with the environment using rewards and penalties.

Explanation:

The model improves its actions based on feedback (reward or punishment).

Example:

Game playing AI (like chess or video games)

13. Explain Network Topology

Definition:

Network topology refers to the structure or arrangement of neurons and layers in a neural network.

Explanation:

It defines how neurons are connected (input → hidden → output).

Example:

Single-layer, multi-layer networks

14. What is Neuron Firing?

Definition:

Neuron firing means a neuron produces an output signal when input exceeds a certain limit I.E . threshold

Explanation:

If input value is strong enough, the neuron activates and sends output.

Example:

If input > threshold → neuron fires

Artificial Neuron

An artificial neuron is a mathematical model inspired by the biological neuron.

It is the basic unit of an Artificial Neural Network (ANN).

ANN is a computational model inspired by the human brain, made of interconnected nodes (neurons) that process information to learn patterns.

Working

1. Inputs are multiplied by weights
2. Bias is added
3. Activation function produces output

Formula

$$Y=f(\sum(wix_i)+b)$$

3. List Components of ANN

ANN stands for Artificial Neural Network.

Main Components

1. Input Layer

- Receives input data
- Example: image pixels, sensor values

2. Hidden Layer

- Performs calculations and feature extraction
- Can be one or multiple layers

3. Output Layer

- Produces final result
- Example: prediction or classification

4. Weights

Weights decide the importance of each input.

Example

If predicting marks:

- Study hours may have high weight
- Playing time may have low weight

5. Bias

- Bias helps shift the activation function.
- It helps the model adjust the output properly instead of always starting from zero."

6. Activation Function

- Decides whether neuron should activate or not

7. Loss Function

- Measures error in prediction

8. Optimizer

- Updates weights to reduce error

Activation Function (Viva Answer)

Definition:

An activation function is a mathematical function used in a neuron to decide whether the neuron should produce output or not.

Explanation:

After calculating weighted sum ($WX + b$), the activation function converts it into the final output.

It introduces non-linearity, which helps the neural network learn complex patterns.

Simple Line to Say in Viva:

“It decides whether a neuron should activate and helps the network learn complex relationships.”

1. Linear Activation Function

Definition:

A function where output is directly proportional to input.

Formula:

$$f(x) = x$$

Explanation:

Simple and easy

No non-linearity

Used in regression problems

Limitation:

Cannot solve complex problems

2. Sigmoid Function

Sigmoid converts input values into values between 0 and 1.

Formula:

$$f(x) = 1 / (1 + e^{-x})$$

It is used in binary classification

Used for probability outputs

And smooth curves

Vanishing gradient problem.

SLOW TRAINING

3. Tanh Function

Tanh converts values between -1 and 1.

ADV:

Zero centered output and better than sigmoid in many cases

Still suffers from the vanishing...problem but less than sigmoid

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

S SHAPED CURVE

4. ReLU (Rectified Linear Unit)

ReLU outputs zero for negative input and same value for positive input.

$$f(x) = \max(0, x)$$

Fast computation

Widely used activation function

Reduces vanishing gradient problem

Used in Deep learning and CNNs.

5. Softmax Function

It converts the output into the probability values

Used in multiclass classification

For example : classifying the images into the cat,dog , man etc.

“Threshold function is an activation function that produces output 1 or 0 depending on whether the input crosses a fixed threshold value.” .

Simple Meaning

- If input is greater than the threshold → output is **1**
- Otherwise → output is **0**

Formula:

$$f(x) = \begin{cases} 1, & x \geq \theta \\ 0, & x < \theta \end{cases}$$

Where:

- x = input
- θ = threshold value

Difference Between Linear and Non-Linear Activation

Linear Activation	Non-Linear Activation
Output proportional to input	Output not proportional
Cannot learn complex patterns	Learns complex patterns
Simple problems only	Complex deep learning tasks
Used in regression	Used in CNN, RNN, ANN

8. What is Perceptron?

A perceptron is the simplest type of artificial neural network developed by Frank Rosenblatt.

It is used for binary classification problems.

Example

- Spam or Not Spam
- Yes or No

Structure of Perceptron

- Inputs
- Weights
- Bias
- Activation Function
- Output

Working

1. Multiply inputs with weights
2. Add bias
3. Apply activation function

4. Generate output

Formula

$$Y=f(\sum w_ix_i+b)$$

Example

If output > threshold → class 1

Else → class 0

9. Limitations of Perceptron

1. Solves Only Linear Problems

Perceptron can only classify linearly separable data.

Example

Cannot solve XOR problem.

2. Single Layer

Basic perceptron has no hidden layers.

3. Limited Learning Ability

Cannot learn complex patterns.

4. Binary Output Only

Mainly used for two-class problems.

5. Low Accuracy for Complex Data

Not suitable for image recognition or NLP tasks.

16. What is Normalization?

Definition:

Normalization is the process of scaling data values to a fixed range, usually 0 to 1.

Explanation:

It helps in making all input features comparable.

Example:

Converting marks from range 0–100 to 0–1

17. Importance of Data Preprocessing

Definition:

Data preprocessing is the process of cleaning and preparing data before training a model.

Explanation (Why important):

- Removes noise and errors
- Improves accuracy
- Makes training faster

Example:

Removing missing values, scaling data

18. Define Feature Scaling

Definition:

Feature scaling is the method of bringing all input features to the same scale.

Explanation:

Prevents one feature from dominating others.

Example:

Age (0–100) and salary (0–100000) → scaled to similar range

19. Explain ANN Architecture Types

“ANN architecture types refer to different structures of artificial neural networks based on how data flows in the network.

1. Feedforward Neural Network – Data flows only in one direction from input to output and it has no loops.
2. Recurrent Neural Network or RNN – It has memory and is mainly used for sequence data like text or speech.
3. Convolutional Neural Network or CNN – It is mainly used for image processing and pattern detection such as edges and shapes.”

20. Real-world Applications of ANN

Definition:

ANN is used to solve real-world problems by learning patterns from data.

Applications:

- Image recognition (face detection)
- Speech recognition (voice assistants)
- Medical diagnosis
- Self-driving cars
- Recommendation systems (Netflix, YouTube)

1. Perceptron Learning Rule

Definition:

It is a rule used to update weights in a perceptron based on error.

Explanation:

If prediction is wrong, weights are adjusted to reduce error.

Formula (simple):

$$W = W + \eta(\text{target} - \text{output})X$$

2. Delta Rule

Definition:

A learning rule used to minimize error by adjusting weights using gradient descent.

Explanation:

It updates weights based on the difference between actual and predicted output.

Perceptron Rule	Delta Rule
Used for classification	Used for regression + classification
Updates only on wrong output	Updates continuously
Uses step function	Uses differentiable function

4. Convergence

Definition:

Convergence is when the model reaches a point where error becomes minimal and stable.

6. Epoch

Definition:

One complete pass of the entire training dataset through the model.

After one epoch the model shows accuracy, precision and loss function

As the no. of epoch increase the accuracy also increases and the loss function decreases

7. Learning Rate

Definition:

A parameter that controls how much weights are updated during training.

8. Effect of High Learning Rate

- Fast learning
 - May skip optimal solution
 - Can cause instability
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9. Effect of Low Learning Rate

- Slow learning
- More accurate
- Takes more time

10. Loss Function

Definition:

A function that measures the error between predicted and actual output.

GRADIENT DECENT AND ITS TYPES

"Gradient Descent is an optimization algorithm used to reduce error and improve the accuracy of a neural network. It updates the weights in the direction where the error decreases fastest.

Types of Gradient Descent:

1. **"Batch Gradient Descent**
Uses the complete dataset to update weights.
It is more stable and accurate but takes more time."
2. **"Stochastic Gradient Descent (SGD)**
Uses one training example at a time.
It is faster but less stable and accuracy may fluctuate."
3. **"Mini-Batch Gradient Descent**
Uses small batches of data for updating weights.
It provides good stability, better accuracy, and less training time, so it is commonly used."

14. Cost Function Minimization

Definition:

Process of reducing error using optimization methods like gradient descent.

MINIMA

"In Gradient Descent, minima are the points where the error is minimum.

1. **"Global Minima – It is the lowest possible error point in the entire graph and gives the best solution."**
2. **"Local Minima – It is a low error point compared to nearby points, but it is not the lowest in the whole graph."**

Gradient Descent tries to reach the global minima to achieve maximum accuracy."

"Momentum is a technique used in Gradient Descent to speed up learning and reduce oscillations during training.

It uses the previous weight update to continue moving in the correct direction.

Advantages:

- "Faster convergence"
- "Reduces zig-zag movement"
- "Helps reach global minima quickly"

Adaptive Learning Rate (Viva Answer)

Definition:

Adaptive learning rate is a technique where the **learning rate automatically adjusts during training** instead of remaining constant.

Explanation:

In normal training, the learning rate is fixed, but in adaptive methods, it changes based on the learning progress.

- If learning is slow → learning rate increases
- If updates are unstable → learning rate decreases

This helps the model learn **faster and more efficiently**.

"Vanishing Gradient is a problem in deep neural networks where the gradients become very small during backpropagation.

Because of this, the earlier layers learn very slowly or stop learning completely.

It usually happens when using sigmoid or tanh activation functions.

Effects:

- "Slow training"
- "Poor accuracy"
- "Difficulty learning deep patterns"

It can be reduced by using ReLU activation, proper weight initialization, and batch normalization."

1. Define MLP

Definition:

MLP (Multi-Layer Perceptron) is a type of neural network with **input layer, hidden layer(s), and output layer.**

Explanation:

It can learn complex patterns using multiple layers.

2. Role of Hidden Layer

Definition:

Hidden layers are intermediate layers that **process and transform input data.**

Explanation:

They help in extracting important features and learning complex relationships.

3. Forward Propagation

Definition:

Forward propagation is the process of **passing input data through the network to get output.**

4. Backpropagation

Definition:

Backpropagation is the process of **updating weights by propagating error backward.**

5. Steps in Backpropagation

1. Forward pass
2. Calculate error
3. Backward pass (compute gradients)
4. Update weights

Forward propagation is the process of the passing input data through multiple layers in the network to get the o/p

Backpropagation continues until error is not minimised

7. Weight Update Formula

Definition:

Formula used to adjust weights to reduce error.

Formula:

$$W = W - \eta \times \text{gradient}$$

8. Error Signal

Definition:

Difference between actual output and predicted output.

WHAT IS GRADIENT

“Gradient is the rate of change of the loss function with respect to weights.

It helps determine how the weights should be updated to reduce error during training.”

10. Training Cycle

Definition:

Complete process of training model over data.

Steps:

Input → Forward pass → Error → Backpropagation → Update weights

Explain steps in ur own words harsh

11. Overfitting

Definition:

“Overfitting happens when the model learns the training data too well, including noise, and performs poorly on new data. (noise means redundant data)

Because of noisy data it cannot perform well on new data

12. Underfitting

Definition:

Underfitting happens when the model is too simple and less trained

And it cannot learn the pattern from the data properly this results in the poor performance in both training and testing data

13. Techniques to Reduce Overfitting

- Dropout
- Early stopping

14. Dropout

Definition:

“Dropout is a technique used to prevent overfitting by randomly deactivating some neurons during training.

15. Early Stopping

Early Stopping is a technique where training is stopped when the validation error starts increasing, to prevent overfitting and improve generalization.”

19. Model Generalization

Definition:

Ability of model to perform well on unseen data.

20. Training vs Testing Data

Training Data	Testing Data
Used to train model	Used to evaluate model
Model learns from it	Model is tested on it

12. Why Sigmoid Causes Vanishing Gradient

Because in extreme values, sigmoid output becomes flat $\rightarrow$ gradients $\approx 0 \rightarrow$ learning slows.

While updating weights the gradient becomes 0 gets stuck

Go to the chat for remaining questions like DNN CNN AND RNN

BAM

Definition:

Bidirectional Associative Memory (BAM) is a type of neural network that stores and recalls pairs of patterns, and works in both directions (forward and backward).

Explanation:

- BAM is a two-layer network
- It connects two sets of neurons (Layer X and Layer Y)
- If you give input from one side, it can recall the associated output from the other side
- It also works in reverse (output $\rightarrow$ input)

Working (Simple):

- Input pattern is given to one layer
- Network recalls the corresponding pair from memory
- Signals move in both directions until stable output is reached

ART

Definition:

ART (Adaptive Resonance Theory) is a type of neural network used for pattern recognition and clustering, which can learn new data without forgetting old data.

Explanation:

ART solves a major problem called stability–plasticity problem:

- Stability $\rightarrow$ remember old patterns
- Plasticity $\rightarrow$ learn new patterns

It checks whether a new input matches existing categories:

- If yes $\rightarrow$ update existing category
- If no $\rightarrow$ create a new category

Key Idea:

It uses a vigilance parameter (threshold) to decide how similar input should be to match a category.

Working (Simple):

1. Input pattern is given
2. Compared with stored patterns
3. If match is good → learning occurs
4. If not → new category is created

HOPFIELD NN

Hopfield Network is a type of recurrent neural network used as associative memory to store and recall patterns.

Explanation:

It consists of neurons that are fully connected with each other

Each neuron acts as both input and output

The network stores patterns and retrieves them for partial or noisy input

It works by minimizing an energy function

#Working (Simple):

Store patterns in the network

Give a partial or noisy input

Network updates neurons step by step

Finally, it settles to the closest stored pattern

Hebbian Learning Rule

Definition:

Hebbian learning states:

"Neurons that fire together wire together."

Formula:

$$W = \sum (x \times x^T)$$

where:

x = pattern vector

Advantages of Hopfield Network

1. Simple associative memory
 2. Noise tolerance
 3. Pattern completion
 4. Fast recall process
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Applications of Hopfield Network

1. Pattern Recognition
 2. Image Restoration
 3. Error Correction
 4. Associative Memory
 5. Optimization Problems
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Logistic Regression

Definition:

Logistic Regression is a classification algorithm used for binary classification problems.

It predicts probability values between:

0 and 1

TensorFlow

Definition:

TensorFlow is an open-source machine learning and deep learning framework developed by Google. It is used to build, train, and deploy neural network models.

PyTorch

PyTorch

Definition:

PyTorch is an open-source deep learning framework developed by Meta (Facebook). It is mainly used for research, neural networks, and AI model development.

Keras

Definition:

Keras is a high-level deep learning API used for building and training neural network models easily. It runs on top of frameworks like TensorFlow.

MNIST Handwritten Character Detection (Viva Answer)

Definition:

MNIST handwritten character detection is a task where a neural network is trained to recognize handwritten digits (0–9) using the MNIST dataset.

Explanation:

- MNIST is a dataset of 28×28 pixel grayscale images of handwritten digits
- The model learns patterns like shapes and strokes
- After training, it can predict which digit (0–9) is in the image

Working (Simple):

1. Input image (28×28 pixels)
2. Pass through neural network (CNN/MLP)
3. Extract features
4. Output predicted digit (0–9)

Key Points:

- Dataset size: ~60,000 training + 10,000 testing images
- Commonly used for beginners in deep learning
- Accuracy can reach >95% easily

Example:

Input image → looks like “5”

Model output → 5